

Ethics in Artificial Intelligence

- Provide a comprehensive introduction to the ethics in artificial intelligence

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3381 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3381.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Artificial Intelligence And Engineering
Course code	3381
Course title	Ethics in Artificial Intelligence
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:3 T:0 P:0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Provide a comprehensive introduction to the ethics in artificial intelligence

Course outcomes

CO	Official outcome	Study level
CO1	Outline the impact of AI 12 Understanding Recognise and address ethical issues in AI and 11 Understanding	See official syllabus
CO2	understand bias mitigation strategies Demonstrate the ability to address ethical concerns, 10 Understanding	See official syllabus
CO3	ensure transparency, and mitigate biases and harms associated with AI systems and datasets.	See official syllabus
CO4	Understand ethical challenges and opportunities in AI 10 Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2 2
CO2	CO2 2 2
CO3	CO3 2 2
CO4	CO4 2 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Outline the impact of AI	6	As specified
Module 2	strategies	4	As specified
Module 3	and mitigate biases and harms associated with AI systems and datasets.	3	As specified
Module 4	Understand ethical challenges and opportunities in AI	4	As specified

MODULE 1

Outline the impact of AI

This module is presented from the official syllabus scope for Ethics in Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the impact of AI
- Official duration: As specified
- Official topic entries captured: 6

- The full source excerpt is preserved below for coverage checking.

Define the morality and ethics in AI 2 Understanding.

Define the morality and ethics in AI 2 Understanding. is an official topic in Module 1 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define the morality and ethics in AI 2 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define the morality and ethics in ai 2 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define the morality and ethics in AI 2 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Define the morality and ethics in AI 2

Understanding. definition/meaning .
 , steps, application . Technical terms
English- .

Outline the impact of AI on society 2 Understanding.

Outline the impact of AI on society 2 Understanding. is an official topic in Module 1 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the impact of AI on society 2 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the impact of ai on society 2 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the impact of AI on society 2 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Outline the impact of AI on society 2

Understanding. definition/meaning .
 , steps, application . Technical terms
English- .

Outline the impact of AI on human psychology 2 Understanding.

Outline the impact of AI on human psychology 2 Understanding. is an official topic in Module 1 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the impact of AI on human psychology 2 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the impact of ai on human psychology 2 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the impact of AI on human psychology 2 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- Outline the impact of AI on human psychology
2 Understanding. definition/meaning .
 , steps, application . Technical terms
English- .

Outline the impact of AI on legal system 2 Understanding. Outline the impact of AI on environment & the

Outline the impact of AI on legal system 2 Understanding. Outline the impact of AI on environment & the is an official topic in Module 1 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the impact of AI on legal system 2 Understanding. Outline the impact of AI on environment & the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the impact of ai on legal system 2 understanding. outline the impact of ai on environment & the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the impact of AI on legal system 2 Understanding. Outline the impact of AI on environment & the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Outline the impact of AI on legal system 2 Understanding. Outline the impact of AI on environment & the definition/meaning . steps, application . Technical terms English-

2 Understanding planet

2 Understanding planet is an official topic in Module 1 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding planet using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding planet should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding planet means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഉറപ്പ് 2 Understanding planet ഗ്രഹത്തിന്റെ നിർവ്വചനം definition/meaning ഗ്രഹം എന്നതിന്റെ അർത്ഥം. ഗ്രഹങ്ങൾ ഉണ്ടാകുന്ന ഘട്ടങ്ങൾ, steps, application ഗ്രഹങ്ങൾ എങ്ങനെ പ്രയോജനപ്പെടുന്നു. Technical terms English-ൽ പദങ്ങൾ നൽകിയിരിക്കുന്നു.

Outline the impact of AI on trust 2 Understanding Definition of morality and ethics in AI- Impact on society-Impact on human psychology- Impact on the legal system -Impact on the environment and the planet-Impact on trust. Recognise and address ethical issues in AI and understand bias mitigation

Outline the impact of AI on trust 2 Understanding Definition of morality and ethics in AI- Impact on society-Impact on human psychology- Impact on the legal system -Impact on the environment and the planet-Impact on trust. Recognise and address ethical issues in AI and understand bias mitigation is an official topic in Module 1 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the impact of AI on trust 2 Understanding Definition of morality and ethics in AI- Impact on society-Impact on human psychology- Impact on the legal system -Impact on the environment and the planet-Impact on trust. Recognise and address ethical issues in AI and understand bias mitigation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the impact of ai on trust 2 understanding definition of morality and ethics in ai- impact on society-impact on human psychology- impact on the legal system -impact on the environment and the planet-impact on trust. recognise and address ethical issues in ai and understand bias mitigation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Outline the impact of AI on trust 2 Understanding Definition of morality and ethics in AI- Impact on society-Impact on human psychology- Impact on the legal system -Impact on the environment and the planet-Impact on trust. Recognise and address ethical issues in AI and understand bias mitigation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓരോ Outline the impact of AI on trust 2 Understanding Definition of morality and ethics in AI- Impact on society-Impact on human psychology- Impact on the legal system -Impact on the environment and the planet-Impact on trust. Recognise and address ethical issues in AI and understand bias mitigation ഓരോരോരോരോരോരോരോരോരോ definition/meaning ഓരോരോരോരോരോരോരോ. ഓരോരോരോ ഓരോരോരോ ഓരോരോരോ, steps, application ഓരോരോരോ ഓരോരോരോരോരോരോ. Technical terms English-ഓ ഓരോരോരോരോരോരോരോരോ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Outline the impact of AI

M1.01	Define the morality and ethics in AI Understanding.	2
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M1.02	Outline the impact of AI on society Understanding.	2
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M1.03	Outline the impact of AI on human psychology Understanding.	2
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M1.04	Outline the impact of AI on legal system Understanding.	2
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M1.05	Outline the impact of AI on environment & the planet Understanding	2
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M1.06	Outline the impact of AI on trust Understanding	2
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Contents:

Definition of morality and ethics in AI- Impact on society-Impact on human psychology-

Impact on the legal system -Impact on the environment and the planet-Impact on trust.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

strategies

This module is presented from the official syllabus scope for Ethics in Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: strategies
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Understand International Ethical Initiatives

Understand International Ethical Initiatives is an official topic in Module 2 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand International Ethical Initiatives using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand international ethical initiatives should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand International Ethical Initiatives means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇന്റർനാഷണൽ എഥിക്കൽ ഇനീഷ്യേറ്റീവ്സ്

ഇന്റർനാഷണൽ എഥിക്സ് definition/meaning ഇന്റർനാഷണൽ എഥിക്സ്. ഇന്റർനാഷണൽ എഥിക്സ്, steps, application ഇന്റർനാഷണൽ എഥിക്സ്. Technical terms English-ഇന്റർനാഷണൽ എഥിക്സ് ഇന്റർനാഷണൽ എഥിക്സ്.

Identify Ethical Harms and Concerns

Identify Ethical Harms and Concerns is an official topic in Module 2 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify Ethical Harms and Concerns using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify ethical harms and concerns should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify Ethical Harms and Concerns means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നിരിക്കുന്നതെന്തെന്ന് തിരിച്ചറിയുക

എന്നിരിക്കുന്നതെന്തെന്ന് തിരിച്ചറിയുക definition/meaning എന്നിരിക്കുന്നതെന്തെന്ന്. എന്നിരിക്കുന്നതെന്തെന്ന്, steps, application എന്നിരിക്കുന്നതെന്തെന്ന് എന്നിരിക്കുന്നതെന്തെന്ന്. Technical terms English-എന്നിരിക്കുന്നതെന്തെന്ന് എന്നിരിക്കുന്നതെന്തെന്ന്.

Analyze Bias in AI Systems

Analyze Bias in AI Systems is an official topic in Module 2 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Analyze Bias in AI Systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, analyze bias in ai systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Analyze Bias in AI Systems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Analyze Bias in AI Systems
 definition/meaning . . . , steps, application Technical terms English-
 .

Summarize Bias Mitigation Techniques 3 Understanding Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms Demonstrate the ability to address ethical concerns, ensure transparency,

Summarize Bias Mitigation Techniques 3 Understanding Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms Demonstrate the ability to address ethical concerns, ensure transparency, is an official topic in Module 2 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Bias Mitigation Techniques 3 Understanding Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms Demonstrate the ability to address ethical concerns, ensure transparency, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize bias mitigation techniques 3 understanding ethical initiatives: international ethical initiatives-ethical harms and concerns-case study: healthcare robots, autonomous vehicles understanding and measuring bias, algorithms for bias mitigation- data bias, preprocessing, in-processing, post-processing algorithms demonstrate the ability to address ethical concerns, ensure transparency, should be

discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Bias Mitigation Techniques 3 Understanding Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms Demonstrate the ability to address ethical concerns, ensure transparency, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Summarize Bias Mitigation Techniques 3 Understanding Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms Demonstrate the ability to address ethical concerns, ensure transparency, definition/meaning . . . , steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

strategies

M2.01	Understand International Ethical Initiatives	3
Understanding		
M2.02	Identify Ethical Harms and Concerns	3
Understanding		
M2.03	Analyze Bias in AI Systems	2
Understanding		
M2.04	Summarize Bias Mitigation Techniques	3
Understanding		
	Series Test – I	1

Contents:

Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-

Case study:

healthcare robots, Autonomous Vehicles

Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms

Demonstrate the ability to address ethical concerns, ensure transparency,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

and mitigate biases and harms associated with AI systems and datasets.

This module is presented from the official syllabus scope for Ethics in Artificial Intelligence. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and mitigate biases and harms associated with AI systems and datasets.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Understand transparency in autonomous systems. 2 Understanding Understand the role of ontological standards in

Understand transparency in autonomous systems. 2 Understanding Understand the role of ontological standards in is an official topic in Module 3 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand transparency in autonomous systems. 2 Understanding Understand the role of ontological standards in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand transparency in autonomous systems. 2 understanding understand the role of ontological standards in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand transparency in autonomous systems. 2 Understanding Understand the role of ontological standards in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓട്ടോമേറ്റഡ് സിസ്റ്റത്തിന്റെ പ്രവർത്തനം. 2 Understanding Understand the role of ontological standards in ഓട്ടോമേറ്റഡ് സിസ്റ്റത്തിന്റെ പ്രവർത്തനം. ഓട്ടോമേറ്റഡ് സിസ്റ്റത്തിന്റെ പ്രവർത്തനം, steps, application ഓട്ടോമേറ്റഡ് സിസ്റ്റത്തിന്റെ പ്രവർത്തനം. Technical terms English-ഓട്ടോമേറ്റഡ് സിസ്റ്റത്തിന്റെ പ്രവർത്തനം.

ethically driven robotics.

ethically driven robotics. is an official topic in Module 3 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ethically driven robotics. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ethically driven robotics. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ethically driven robotics. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എഥിക്കലി ഡ്രൈവൻ റോബട്ടിക്സ്. റോബട്ടിക്സ് ന്റെ ഡിഫിനിഷൻ/മീനിംഗ്. റോബട്ടിക്സ് ന്റെ ഡിഫിനിഷൻ, steps, application റോബട്ടിക്സ് ന്റെ ഡിഫിനിഷൻ. Technical terms English-എഥിക്കലി ഡ്രൈവൻ റോബട്ടിക്സ്.

Identify challenges to government adoption of AI 2 Understanding Understand the role of datasets and mitigate 2 Understanding M3.05 associated harms. AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design - Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms.

Identify challenges to government adoption of AI 2 Understanding Understand the role of datasets and mitigate 2 Understanding M3.05 associated harms. AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design - Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms. is an official topic in Module 3 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify challenges to government adoption of AI 2 Understanding Understand the role of datasets and mitigate 2 Understanding M3.05 associated harms. AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design - Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify challenges to government adoption of ai 2 understanding understand the role of datasets and mitigate 2 understanding m3.05 associated harms. ai standards and regulation model process for addressing ethical concerns during system design – transparency of autonomous systems-data privacy process- algorithmic bias considerations, ontological standard for ethically driven robotics and automation systems challenges to government adoption of ai datasets in different domains, roles of datasets, harms associated with data, mitigating harms. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify challenges to government adoption of AI 2 Understanding Understand the role of datasets and mitigate 2 Understanding M3.05 associated harms. AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design – Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-Identify challenges to government adoption of AI 2 Understanding Understand the role of datasets and mitigate 2 Understanding M3.05 associated harms. AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design – Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms. definition/meaning . steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

and mitigate biases and harms associated with AI systems and datasets.

Understanding
M3.01 Apply ethical considerations throughout AI system design. 2

M3.02 Understand transparency in autonomous systems. 2
Understanding

Understanding
M3.03 Understand the role of ontological standards in ethically driven robotics. 2

M3.04 Identify challenges to government adoption of AI 2
Understanding

Understanding
M3.05 Understand the role of datasets and mitigate associated harms. 2

Contents:

AI STANDARDS AND REGULATION

Model Process for Addressing Ethical Concerns during System Design – Transparency of

Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological

Standard for Ethically Driven Robotics and Automation Systems

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Understand ethical challenges and opportunities in AI

This module is presented from the official syllabus scope for Ethics in Artificial Intelligence. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand ethical challenges and opportunities in AI
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Remembering AI.

3 Remembering AI. is an official topic in Module 4 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering AI. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering ai. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering AI. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Remembering AI. ഓർമ്മപ്പെടുത്തൽ ഓർമ്മ
definition/meaning ഓർമ്മപ്പെടുത്തൽ. ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ, steps, application
ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ. Technical terms English-ഓർമ്മപ്പെടുത്തൽ ഓർമ്മപ്പെടുത്തൽ.

Analyze societal implications of AI applications 3 Understanding Understand decision-making roles across

Analyze societal implications of AI applications 3 Understanding Understand decision-making roles across is an official topic in Module 4 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Analyze societal implications of AI applications 3 Understanding Understand decision-making roles across using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, analyze societal implications of ai applications 3 understanding understand decision-making roles across should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Analyze societal implications of AI applications 3 Understanding Understand decision-making roles across means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നു വിശദീകരിക്കുക. Analyze societal implications of AI applications
3 Understanding Understand decision-making roles across വിവിധ മേഖലകളിൽ
definition/meaning വിശദീകരിക്കുക. വിശദീകരിക്കുക വിശദീകരിക്കുക, steps, application
വിശദീകരിക്കുക വിശദീകരിക്കുക വിശദീകരിക്കുക. Technical terms English-ൽ വിശദീകരിക്കുക വിശദീകരിക്കുക.

2 Understanding industries in the context of AI. Understand national and international strategies

2 Understanding industries in the context of AI. Understand national and international strategies is an official topic in Module 4 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding industries in the context of AI. Understand national and international strategies using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding industries in the context of ai. understand national and international strategies should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding industries in the context of AI. Understand national and international strategies means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഏകദേശം 2 Understanding industries in the context of AI. Understand national and international strategies ഏകദേശം 2 definition/meaning ഏകദേശം 2. ഏകദേശം 2 ഏകദേശം 2, steps, application ഏകദേശം 2 ഏകദേശം 2. Technical terms English-ഏകദേശം 2 ഏകദേശം 2.

2 Understanding for governing AI. AI AND ETHICS- CHALLENGES AND OPPORTUNITIES Challenges - Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI. Text / Reference T/R Book Title/Author Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp T1 and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 - March 2020 Trisha Mahoney, Kush R. Varshney & Michael Hind, AI Fairness - How to Measure R1 and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020 R2 Ethics of Artificial Intelligence" Vincent C. Müller ,MIT Press 2020 Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm

2 Understanding for governing AI. AI AND ETHICS- CHALLENGES AND OPPORTUNITIES Challenges - Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI. Text / Reference T/R Book Title/Author Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp T1 and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 - March 2020 Trisha Mahoney, Kush R. Varshney & Michael Hind, AI Fairness - How to Measure R1 and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020 R2 Ethics of Artificial Intelligence" Vincent C. Müller ,MIT Press 2020 Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm is an official topic in Module 4 of Ethics in Artificial Intelligence. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding for governing AI. AI AND ETHICS- CHALLENGES AND OPPORTUNITIES Challenges – Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI. Text / Reference T/R Book Title/Author Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp T1 and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 – March 2020 Trisha Mahoney, Kush R. Varshney & Michael Hind, AI Fairness - How to Measure R1 and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020 R2 Ethics of Artificial Intelligence" Vincent C. Müller ,MIT Press 2020 Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding for governing ai. ai and ethics- challenges and opportunities challenges – opportunities- ethical issues in artificial intelligence- societal issues concerning the application of artificial intelligence in medicine- decision-making role in industries-national and international strategies on ai. text / reference t/r book title/author y. eleanor bird, jasmin fox-skelly, nicola jenner, ruth larbey, emma weitkamp t1 and alan winfield ,the ethics of artificial intelligence: issues and initiatives, eprs | european parliamentary research service scientific foresight unit (stoa) pe 634.452 – march 2020 trisha mahoney, kush r. varshney & michael hind, ai fairness - how to measure r1 and reduce unwanted bias in machine learning, oreilly, 2020 r2 ethics of artificial intelligence" vincent c. müller ,mit press 2020 online resources sl.no website link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding for governing AI. AI AND ETHICS- CHALLENGES AND OPPORTUNITIES Challenges – Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI. Text / Reference T/R Book Title/Author Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp T1 and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 – March 2020 Trisha Mahoney, Kush R. Varshney & Michael Hind,

Aspect	What to prepare
	AI Fairness - How to Measure R1 and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020 R2 Ethics of Artificial Intelligence" Vincent C. Müller ,MIT Press 2020 Online Resources Sl.No Website Link 1 https://nptel.ac.in/courses/106/105/106105214/# 2 https://www.tutorialspoint.com/operating_system/index.htm means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding for governing AI. AI AND ETHICS- CHALLENGES AND OPPORTUNITIES Challenges – Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI. Text / Reference T/R Book Title/Author Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp T1 and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 – March 2020 Trisha Mahoney, Kush R. Varshney & Michael Hind, AI Fairness - How to Measure R1 and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020 R2 Ethics of Artificial Intelligence" Vincent C. Müller ,MIT Press 2020 Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/#> 2 https://www.tutorialspoint.com/operating_system/index.htm definition/meaning . , steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Understand ethical challenges and opportunities in AI

	Identify ethical challenges and opportunities in	
M4.01		3
Remembering	AI.	
M4.02	Analyze societal implications of AI applications	3
Understanding	Understand decision-making roles across	
M4.03		2
Understanding	industries in the context of AI.	
	Understand national and international strategies	
M4.04		2
Understanding	for governing AI.	
	Series Test – II	1

Contents:

AI AND ETHICS- CHALLENGES AND OPPORTUNITIES

Challenges – Opportunities- ethical issues in artificial intelligence- Societal Issues

Concerning the Application of Artificial Intelligence in Medicine- decision-making role in

industries-National and International Strategies on AI.

Text / Reference

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define the morality and ethics in AI 2 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Define the morality and ethics in AI 2 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline the impact of AI on society 2 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the impact of AI on society 2 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline the impact of AI on human psychology 2 Understanding.. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the impact of AI on human psychology 2 Understanding..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline the impact of AI on legal system 2 Understanding. Outline the impact of AI on environment & the. (3 marks)

Answer: Begin with the standard definition or identity of *Outline the impact of AI on legal system 2 Understanding. Outline the impact of AI on environment & the.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain Understand International Ethical Initiatives. (3 marks)

Answer: Begin with the standard definition or identity of *Understand International Ethical Initiatives*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Identify Ethical Harms and Concerns. (3 marks)

Answer: Begin with the standard definition or identity of *Identify Ethical Harms and Concerns*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Analyze Bias in AI Systems. (3 marks)

Answer: Begin with the standard definition or identity of *Analyze Bias in AI Systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Summarize Bias Mitigation Techniques 3 Understanding Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms Demonstrate the ability to address ethical concerns, ensure transparency,. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize Bias Mitigation Techniques 3 Understanding Ethical Initiatives: International ethical initiatives-Ethical harms and concerns-Case study: healthcare robots, Autonomous Vehicles Understanding and measuring Bias, Algorithms for Bias mitigation- Data Bias, Preprocessing, in-processing, post-processing algorithms Demonstrate the ability to address ethical concerns, ensure transparency*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Understand transparency in autonomous systems. 2 Understanding Understand the role of ontological standards in. (3 marks)

Answer: Begin with the standard definition or identity of *Understand transparency in autonomous systems. 2 Understanding Understand the role of ontological standards in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain ethically driven robotics.. (3 marks)

Answer: Begin with the standard definition or identity of *ethically driven robotics*.. State the governing principle, list the key components or stages, and close with one

relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Identify challenges to government adoption of AI 2 Understanding Understand the role of datasets and mitigate 2 Understanding M3.05 associated harms. AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design - Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms.. (3 marks)

Answer: Begin with the standard definition or identity of *Identify challenges to government adoption of AI 2 Understanding Understand the role of datasets and mitigate 2 Understanding M3.05 associated harms. AI STANDARDS AND REGULATION Model Process for Addressing Ethical Concerns during System Design - Transparency of Autonomous Systems-Data Privacy Process- Algorithmic Bias Considerations, Ontological Standard for Ethically Driven Robotics and Automation Systems Challenges to Government Adoption of AI Datasets in different domains, Roles of datasets, harms associated with data, mitigating harms..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Remembering AI.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering AI..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Analyze societal implications of AI applications 3 Understanding Understand decision-making roles across. (3 marks)

Answer: Begin with the standard definition or identity of *Analyze societal implications of AI applications 3 Understanding Understand decision-making roles across*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 2 Understanding industries in the context of AI. Understand national and international strategies. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding industries in the context of AI. Understand national and international strategies*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 2 Understanding for governing AI. AI AND ETHICS- CHALLENGES AND OPPORTUNITIES Challenges - Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI. Text / Reference T/R Book Title/Author Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp T1 and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 - March 2020 Trisha Mahoney, Kush R. Varshney & Michael Hind, AI Fairness - How to Measure R1 and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020 R2 Ethics of Artificial

Intelligence" Vincent C. Müller ,MIT Press 2020 Online Resources Sl.No
Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/# 2>
https://www.tutorialspoint.com/operating_system/index.htm. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Understanding for governing AI. AI AND ETHICS- CHALLENGES AND OPPORTUNITIES Challenges - Opportunities- ethical issues in artificial intelligence- Societal Issues Concerning the Application of Artificial Intelligence in Medicine- decision-making role in industries-National and International Strategies on AI. Text / Reference T/R Book Title/Author Y. Eleanor Bird, Jasmin Fox-Skelly, Nicola Jenner, Ruth Larbey, Emma Weitkamp T1 and Alan Winfield ,The ethics of artificial intelligence: Issues and initiatives, EPRS | European Parliamentary Research Service Scientific Foresight Unit (STOA) PE 634.452 - March 2020 Trisha Mahoney, Kush R. Varshney & Michael Hind, AI Fairness - How to Measure R1 and Reduce Unwanted Bias in Machine Learning, Oreilly, 2020 R2 Ethics of Artificial Intelligence" Vincent C. Müller ,MIT Press 2020 Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/106/105/106105214/# 2> https://www.tutorialspoint.com/operating_system/index.htm. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define the morality and ethics in AI 2 Understanding..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Understand International Ethical Initiatives**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Understand transparency in autonomous systems. 2 Understanding Understand the role of ontological standards in.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Remembering AI..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://nptel.ac.in/courses/106/105/106105214/#>

- https://www.tutorialspoint.com/operating_system/index.htm

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3381. Verify institutional updates against the current official curriculum.